

Task Questions

1. Question 1

Yes, an S3 bucket was created even if no name was specified. The bucket was given a default name, which includes and have the format [stack name]-[logical ID of resources in the template]-[some random letters and numbers], something like this createbucket-s3bucket-<random-string

2. Question 2

It was created in the AWS region in which the Cloud formation create stack command was run.

3. Question 3

Only 2 lines:

S3Bucket:

Type: AWS::S3::Bucket

4. Question 4

Once I reached the parameters tab of the update-café-app stack, I saw the value /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2 for the LatestAmild

5. Question 5

The ARN of the IAM role that grants permission to create and update the update-café-app stack is CustomCFnRole

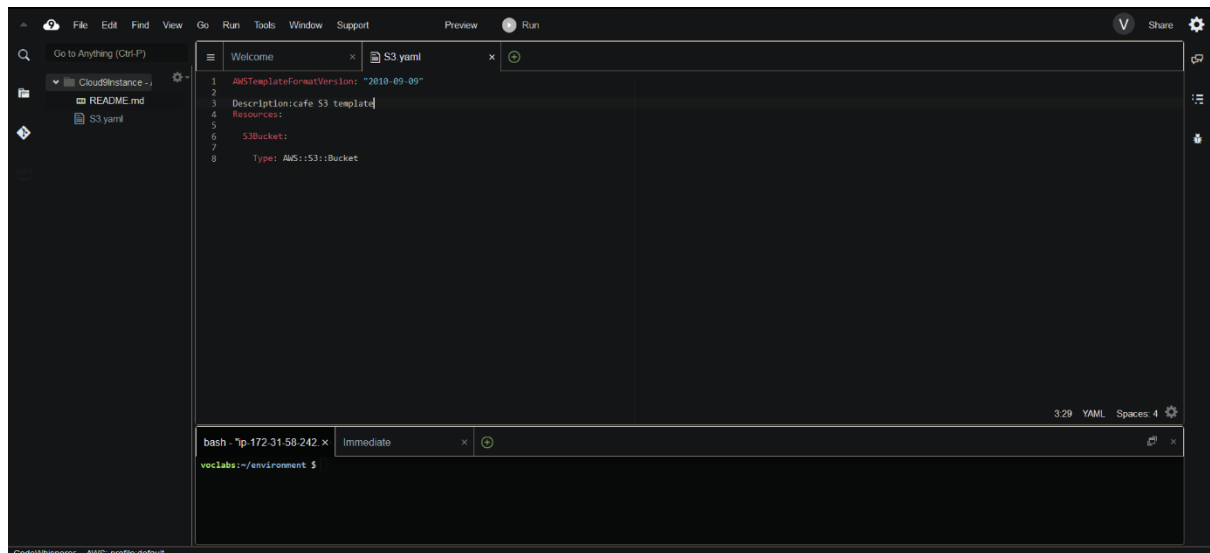
6. Question 6

Details of what lines of which files were altered as part of the commit and what the lines now have.

Module 10 Challenge Lab - Automating Infrastructure Deployment

Task 1: Creating an AWS CloudFormation template from scratch

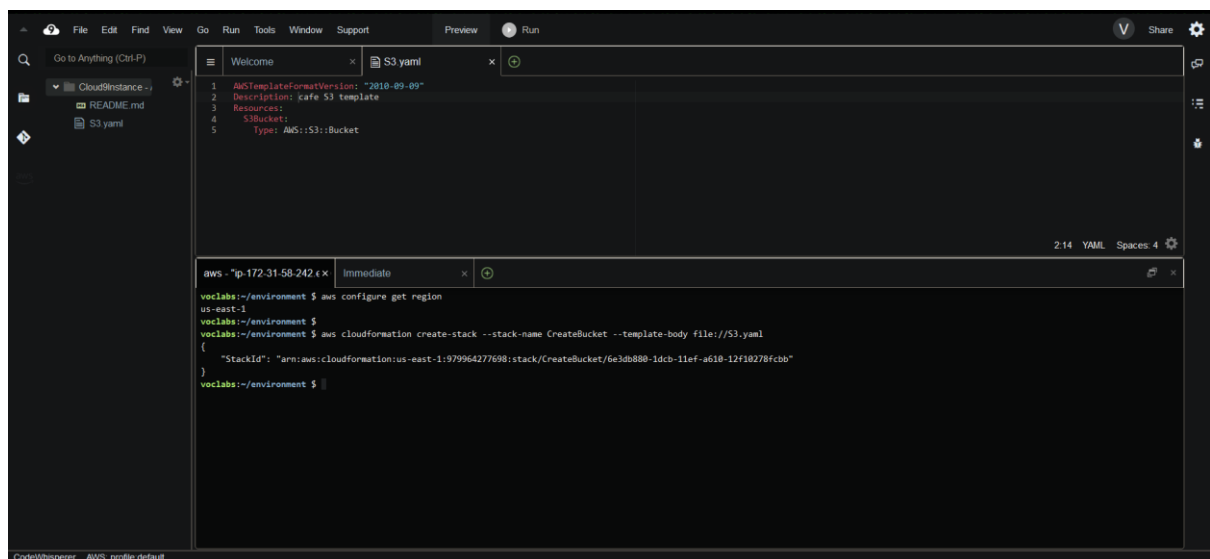
In this task, I create an AWS CloudFormation template to set up an S3 bucket. First, I navigate to AWS Cloud9 and open the IDE. I create a new file named S3.yaml and add the AWS template version and a description.



The screenshot shows the AWS Cloud9 IDE interface. The file explorer on the left shows a project named 'Cloud9Instance-' with files 'README.md' and 'S3.yaml'. The main editor window has two tabs: 'Welcome' and 'S3.yaml'. The 'S3.yaml' file contains the following content:

```
1 AWSTemplateFormatVersion: "2010-09-09"
2
3 Description: kafe S3 template
4 Resources:
5
6   S3Bucket:
7     Type: AWS::S3::Bucket
```

The status bar at the bottom right indicates '3/29 YAML Spaces: 4'. A terminal window at the bottom shows the prompt 'voclabs:~/environment \$'.



The screenshot shows the AWS Cloud9 IDE interface after running several commands. The file explorer on the left shows the project 'Cloud9Instance-' with files 'README.md' and 'S3.yaml'. The main editor window has two tabs: 'Welcome' and 'S3.yaml'. The 'S3.yaml' file contains the following content:

```
1 AWSTemplateFormatVersion: "2010-09-09"
2
3 Description: kafe S3 template
4 Resources:
5   S3Bucket:
6     Type: AWS::S3::Bucket
```

The status bar at the bottom right indicates '2/14 YML Spaces: 4'. A terminal window at the bottom shows the following commands and output:

```
aws - "ip-172-31-58-242.x" Immediate
voclabs:~/environment $ aws configure get region
us-east-1
voclabs:~/environment $
voclabs:~/environment $ aws cloudformation create-stack --stack-name CreateBucket --template-body file://S3.yaml
{
  "StackId": "arn:aws:cloudformation:us-east-1:979964277698:stack/CreateBucket/6e3db880-1dcb-11ef-a610-12f10278fcb"
}
voclabs:~/environment $
```

I then define an S3 bucket resource with the correct indentation. After saving the file, I switch to the Bash terminal and run commands to get the default AWS region and create a CloudFormation stack using my template.

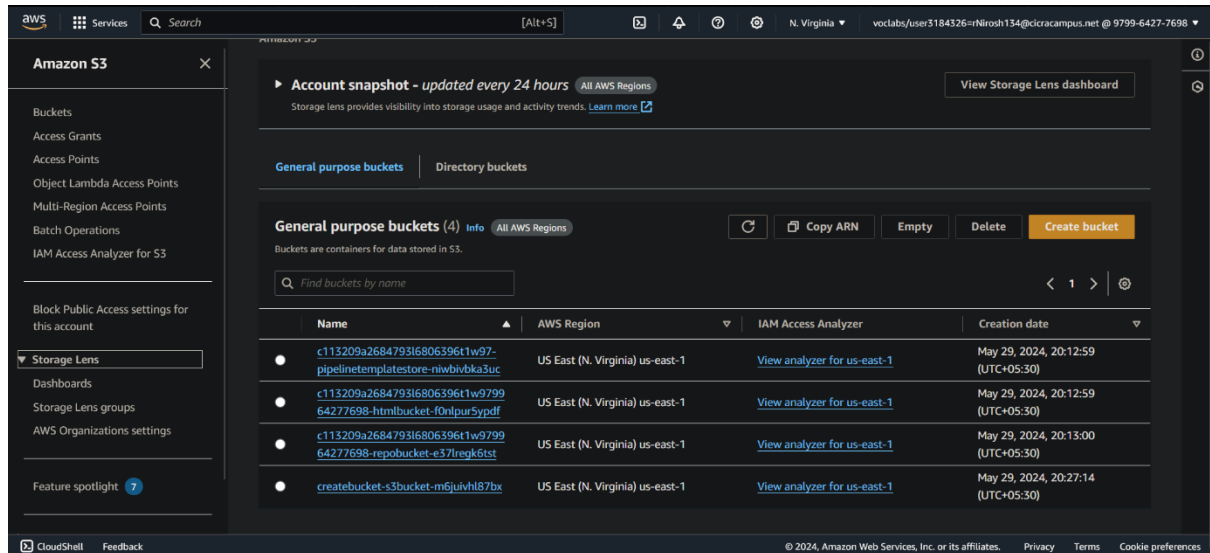
The screenshot shows the AWS CloudFormation console. The left sidebar contains navigation options: Stacks, Stack details, Drifts, StackSets, Exports, Application Composer, IaC generator, Registry, Public extensions, Activated extensions, Publisher, Spotlight, and Feedback. The main content area is titled 'CloudFormation > Stacks > CreateBucket'. It shows a list of stacks with columns for Name, Status, and Last updated. The 'CreateBucket' stack is highlighted, showing a status of 'CREATE_COMPLETE'. The right pane shows the 'Events' tab for the 'CreateBucket' stack, displaying a list of events with columns for Timestamp, Logical ID, Status, and Detailed status. The events show the successful creation of the 'CreateBucket' stack and the 'S3Bucket' resource.

Timestamp	Logical ID	Status	Detailed status
2024-05-29 20:27:34 UTC+0530	CreateBucket	CREATE_COMPLETE	-
2024-05-29 20:27:34 UTC+0530	S3Bucket	CREATE_COMPLETE	-
2024-05-29 20:27:21 UTC+0530	CreateBucket	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE
2024-05-29 20:27:21 UTC+0530	S3Bucket	CREATE_IN_PROGRESS	CONFIGURATION_COMPLETE
2024-05-29 20:27:12	S3Bucket	CREATE_IN_PROGRESS	-

The screenshot shows the AWS CloudFormation console, specifically the 'Template' tab for the 'CreateBucket' stack. The left sidebar is the same as the previous screenshot. The main content area shows the 'Template' tab, which displays the template content. The template is an AWS::S3::Bucket resource named 'S3Bucket'. The right pane shows the 'Template' tab, displaying the template content. The template is an AWS::S3::Bucket resource named 'S3Bucket'. The right pane shows the 'Template' tab, displaying the template content. The template is an AWS::S3::Bucket resource named 'S3Bucket'.

```
AWSTemplateFormatVersion: "2010-09-09"
Description: cafe s3 template
Resources:
  S3Bucket:
    Type: AWS::S3::Bucket
```

If successful, I see JSON output with a StackId. I verify the stack in the AWS Management Console by checking its details and confirm the S3 bucket's creation on the Amazon S3 service page, where it appears with a name like createbucket-s3bucket-<random-string>.



View questions in: [English](#)

Question 1: Was an S3 bucket created, even if you did not specify a name for the bucket? If so, what name was it given?

☐ No, the bucket was not created.

☐ Yes, and it was given a name in the format [logical ID of resource in the template]-[stack name]-[some random letters and numbers]

☒ Yes, and it was given a name in the format [stack name]-[logical ID of resource in the template]-[some random letters and numbers]

☐ Yes, and it was given a name in the format [logical ID of resource in the template]-[stack name]

Question 2: What Region was the bucket created in, and why was it created in this Region?

☐ It was created in the us-east-1 AWS Region, because that is always the default AWS Region for every AWS CLI client unless you specify a different AWS Region when running the command.

☒ It was created in the AWS Region in which the CloudFormation create-stack command was run.

☐ It was created in the ap-northeast-1 AWS Region because that was set as the default region in the AWS CLI client on the Cloud9 instance.

☐ It was created in the eu-west-2 AWS Region because that was set as the default region in the AWS CLI client on the Cloud9 instance.

Question 3: To define an S3 bucket, how many lines of code did you need to enter in the "Resources:" section of the template file?

☒ 2 lines

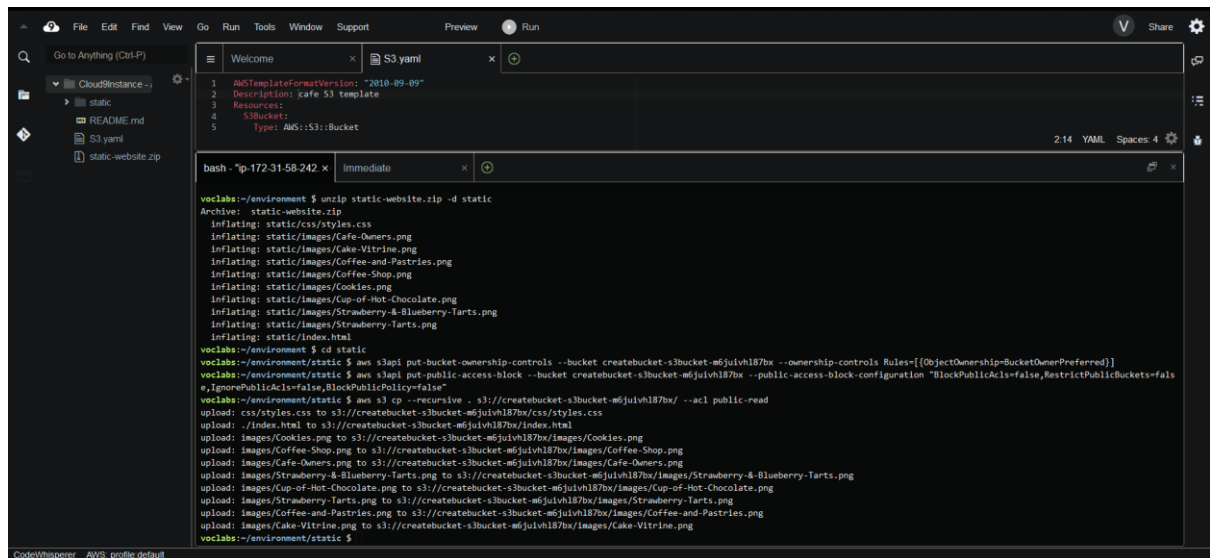
☐ 3 lines

☐ 4 lines

☐ 5 lines

Task 2: Configuring the bucket as a website and updating the stack

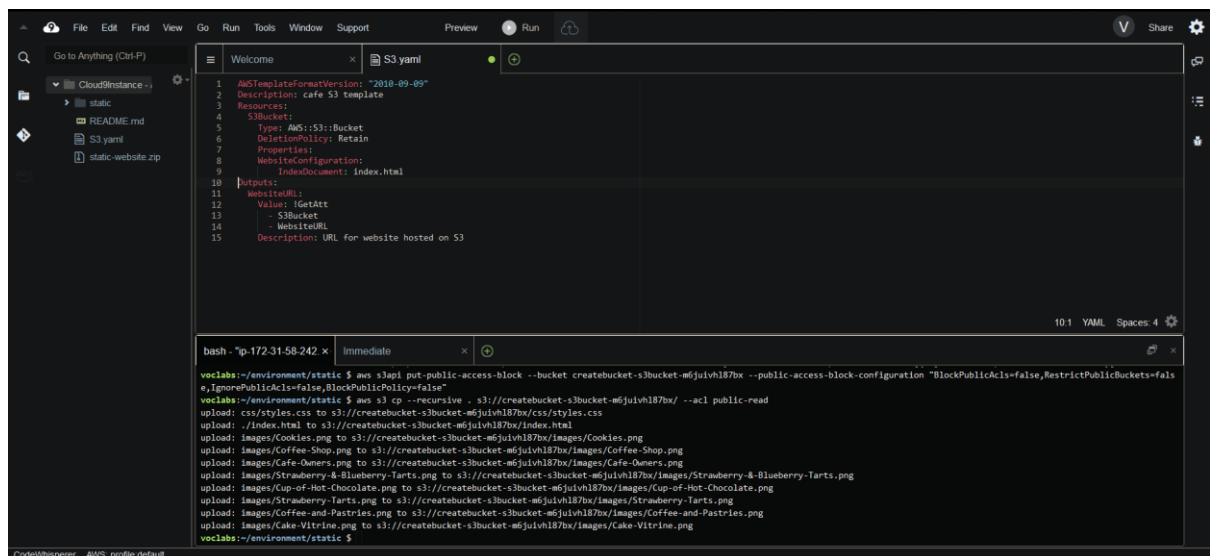
In this task, I update my AWS CloudFormation template to configure an S3 bucket for static website hosting. I start by running commands in the Bash terminal to download, unzip, and upload the website files to the bucket, while also setting ownership controls and public access settings.



```
1 AWSTemplateFormatVersion: "2010-09-09"
2 Description: cafe S3 template
3 Resources:
4   S3Bucket:
5     Type: AWS::S3::Bucket

bash - "ip-172-31-58-242.x" Immediate x

voclabs:~/environment $ unzip static-website.zip -d static
Archive: static-website.zip
  inflating: static/css/styles.css
  inflating: static/images/Cafe-Owners.png
  inflating: static/images/Cake-Vitrine.png
  inflating: static/images/Coffee-and-Pastries.png
  inflating: static/images/Coffee-Shop.png
  inflating: static/images/Cookies.png
  inflating: static/images/Cup-of-Hot-Chocolate.png
  inflating: static/images/Strawberry-&Blueberry-Tarts.png
  inflating: static/images/Strawberry-Tarts.png
  inflating: static/index.html
voclabs:~/environment $ cd static
voclabs:~/environment/static $ aws s3api put-bucket-ownership-controls --bucket createbucket-s3bucket-m6juivh187bx --ownership-controls Rules-[[ObjectOwnership:BucketOwnerPreferred]]
voclabs:~/environment/static $ aws s3api put-public-access-block --bucket createbucket-s3bucket-m6juivh187bx --public-access-block-configuration "BlockPublicAcls=false,RestrictPublicBuckets=false,IgnorePublicAcls=false,BucketPublicPolicy=false"
voclabs:~/environment/static $ aws s3 cp --recursive . s3://createbucket-s3bucket-m6juivh187bx/ --acl public-read
upload: css/styles.css to s3://createbucket-s3bucket-m6juivh187bx/css/styles.css
upload: ./index.html to s3://createbucket-s3bucket-m6juivh187bx/index.html
upload: Images/Cookies.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cookies.png
upload: Images/Coffee-Shop.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Coffee-Shop.png
upload: Images/Cafe-Owners.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cafe-Owners.png
upload: Images/Strawberry-&Blueberry-Tarts.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Strawberry-&Blueberry-Tarts.png
upload: Images/Cup-of-Hot-Chocolate.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cup-of-Hot-Chocolate.png
upload: Images/Strawberry-Tarts.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Strawberry-Tarts.png
upload: Images/Coffee-and-Pastries.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Coffee-and-Pastries.png
upload: Images/Cake-Vitrine.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cake-Vitrine.png
voclabs:~/environment/static $
```

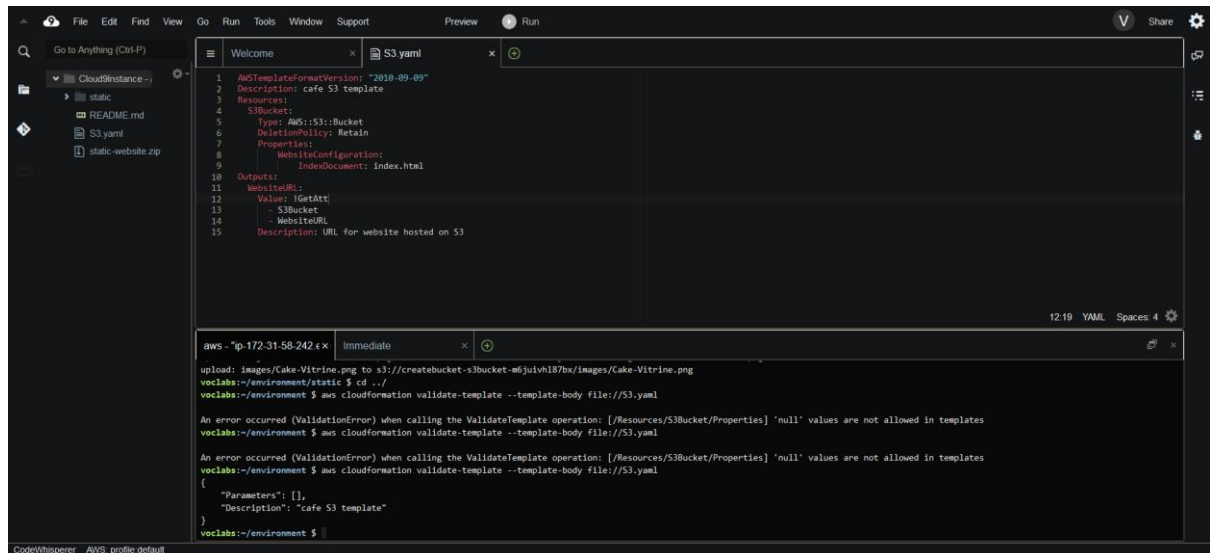


```
1 AWSTemplateFormatVersion: "2010-09-09"
2 Description: cafe S3 template
3 Resources:
4   S3Bucket:
5     Type: AWS::S3::Bucket
6     DeletionPolicy: Retain
7     Properties:
8       WebsiteConfiguration:
9         IndexDocument: index.html
10    Outputs:
11      WebsiteURL:
12        Value: !GetAtt
13          - S3Bucket
14        WebsiteURL
15    Description: URL for website hosted on S3

bash - "ip-172-31-58-242.x" Immediate x

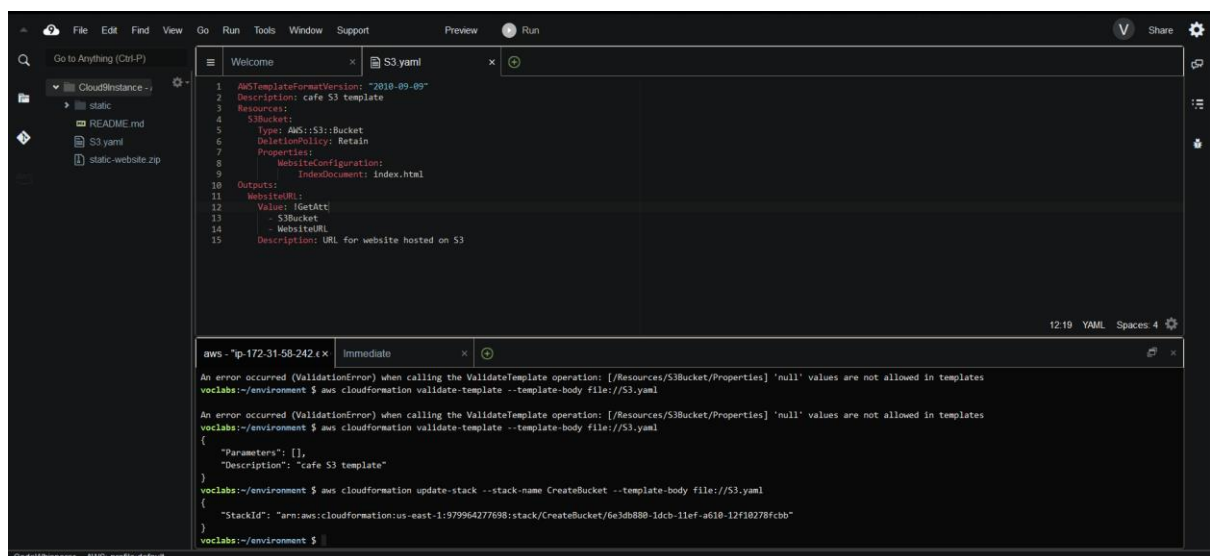
voclabs:~/environment/static $ aws s3api put-public-access-block --bucket createbucket-s3bucket-m6juivh187bx --public-access-block-configuration "BlockPublicAcls=false,RestrictPublicBuckets=false,IgnorePublicAcls=false,BucketPublicPolicy=false"
voclabs:~/environment/static $ aws s3 cp --recursive . s3://createbucket-s3bucket-m6juivh187bx/ --acl public-read
upload: css/styles.css to s3://createbucket-s3bucket-m6juivh187bx/css/styles.css
upload: ./index.html to s3://createbucket-s3bucket-m6juivh187bx/index.html
upload: Images/Cookies.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cookies.png
upload: Images/Coffee-Shop.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Coffee-Shop.png
upload: Images/Cafe-Owners.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cafe-Owners.png
upload: Images/Strawberry-&Blueberry-Tarts.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Strawberry-&Blueberry-Tarts.png
upload: Images/Cup-of-Hot-Chocolate.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cup-of-Hot-Chocolate.png
upload: Images/Strawberry-Tarts.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Strawberry-Tarts.png
upload: Images/Coffee-and-Pastries.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Coffee-and-Pastries.png
upload: Images/Cake-Vitrine.png to s3://createbucket-s3bucket-m6juivh187bx/Images/Cake-Vitrine.png
voclabs:~/environment/static $
```

I then modify my S3.yml template to retain the bucket on deletion, set it up for static website hosting with index.html as the index document, and add an output for the website URL. After saving and validating the template, I update the CloudFormation stack.



```
1 AWSTemplateFormatVersion: "2010-09-09"
2 Description: cafe S3 template
3 Resources:
4   S3Bucket:
5     Type: AWS::S3::Bucket
6     DeletionPolicy: Retain
7     Properties:
8       WebsiteConfiguration:
9         IndexDocument: index.html
10  Outputs:
11    WebsiteURL:
12      Value: !GetAtt
13        - S3Bucket
14      - WebsiteURL
15    Description: URL for website hosted on S3
```

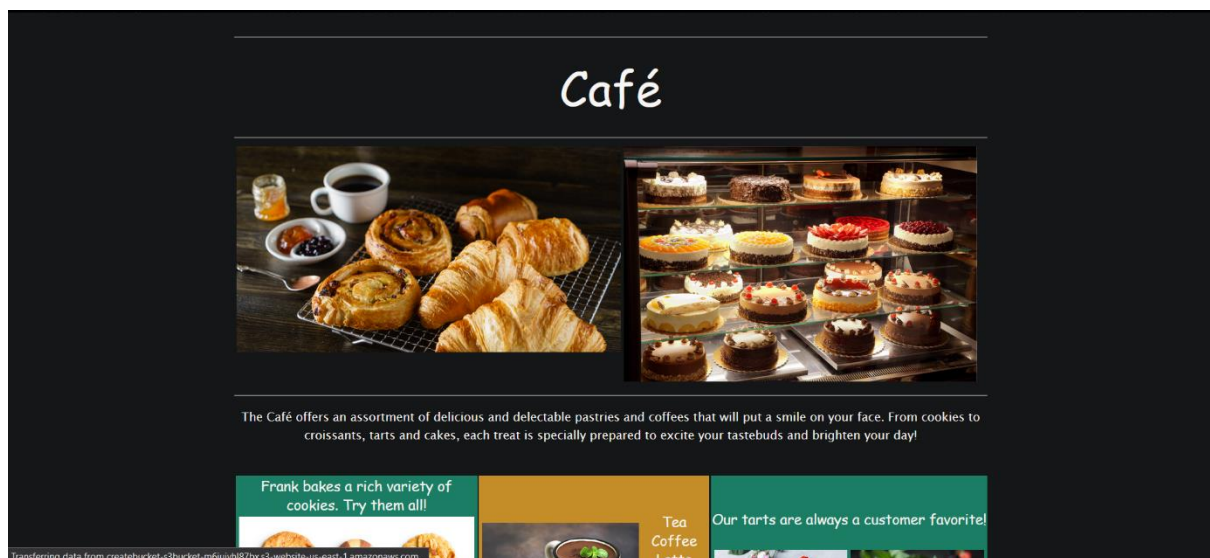
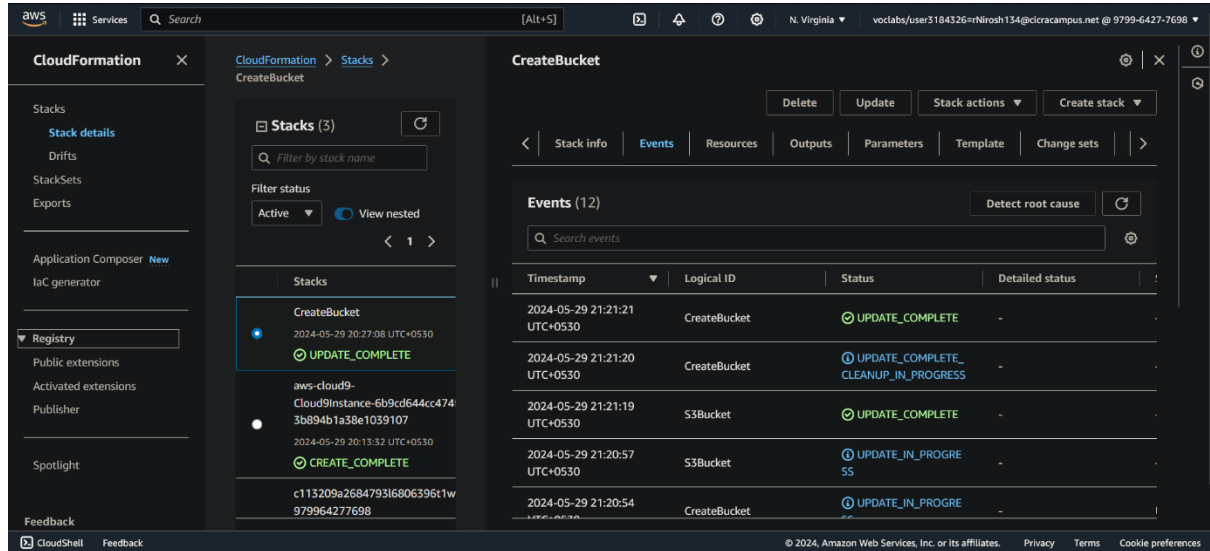
```
aws - "ip-172-31-58-242.c" Immediate
upload: Images/Cake-Vitrine.png to s3://createbucket-s3bucket-m6juivh187bx/images/Cake-Vitrine.png
voclabs:~/environment/static $ cd ../
voclabs:~/environment $ aws cloudformation validate-template --template-body file://S3.yml
An error occurred (ValidationError) when calling the ValidateTemplate operation: [/Resources/S3Bucket/Properties] 'null' values are not allowed in templates
voclabs:~/environment $ aws cloudformation validate-template --template-body file://S3.yml
An error occurred (ValidationError) when calling the ValidateTemplate operation: [/Resources/S3Bucket/Properties] 'null' values are not allowed in templates
voclabs:~/environment $ aws cloudformation validate-template --template-body file://S3.yml
{
  "Parameters": [],
  "Description": "cafe S3 template"
}
voclabs:~/environment $
```



```
1 AWSTemplateFormatVersion: "2010-09-09"
2 Description: cafe S3 template
3 Resources:
4   S3Bucket:
5     Type: AWS::S3::Bucket
6     DeletionPolicy: Retain
7     Properties:
8       WebsiteConfiguration:
9         IndexDocument: index.html
10  Outputs:
11    WebsiteURL:
12      Value: !GetAtt
13        - S3Bucket
14      - WebsiteURL
15    Description: URL for website hosted on S3
```

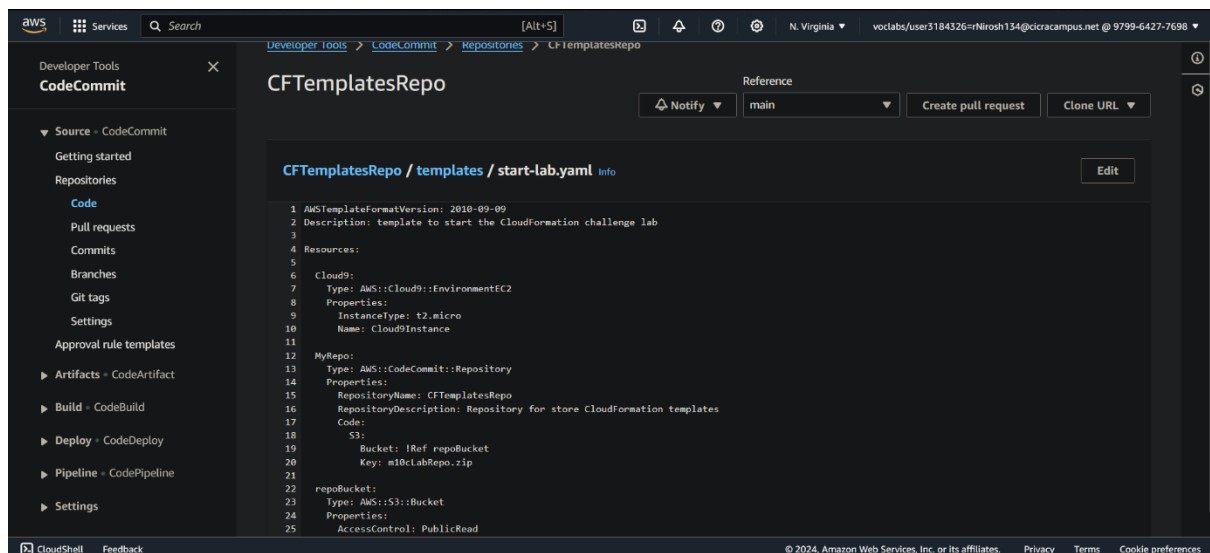
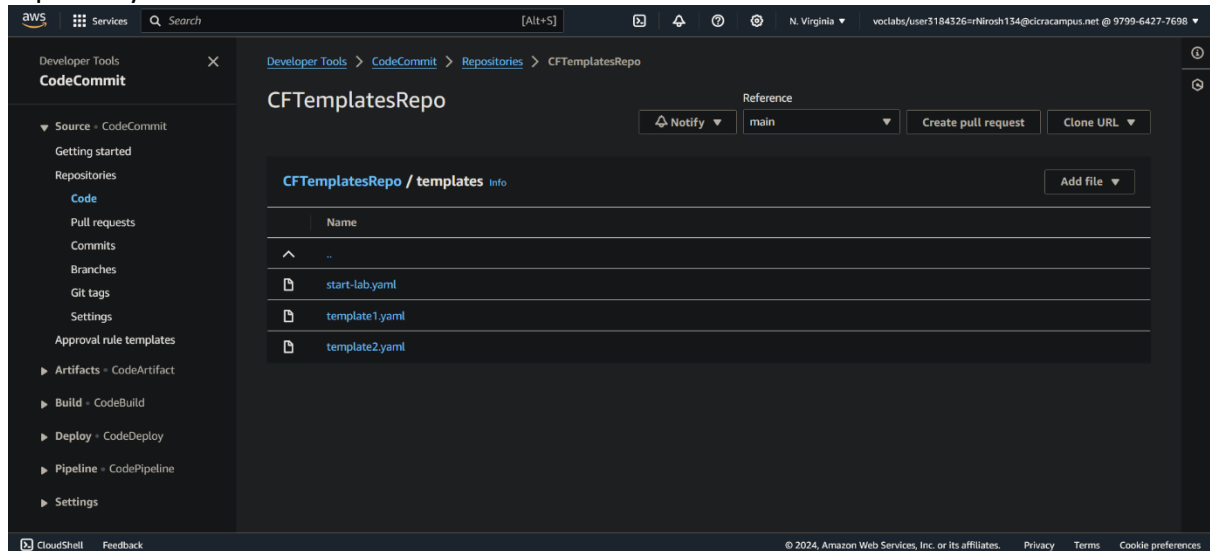
```
aws - "ip-172-31-58-242.c" Immediate
An error occurred (ValidationError) when calling the ValidateTemplate operation: [/Resources/S3Bucket/Properties] 'null' values are not allowed in templates
voclabs:~/environment $ aws cloudformation validate-template --template-body file://S3.yml
An error occurred (ValidationError) when calling the ValidateTemplate operation: [/Resources/S3Bucket/Properties] 'null' values are not allowed in templates
voclabs:~/environment $ aws cloudformation validate-template --template-body file://S3.yml
{
  "Parameters": [],
  "Description": "cafe S3 template"
}
voclabs:~/environment $ aws cloudformation update-stack --stack-name CreateBucket --template-body file://S3.yml
{
  "StackId": "arn:aws:cloudformation:us-east-1:979964277698:stack/CreateBucket/6e3db880-1dc6-11ef-a610-12f18278fcb6"
}
voclabs:~/environment $
```

I confirm the update's success in the AWS Management Console and verify the website by checking the stack's Outputs tab for the URL and ensuring the site loads correctly.

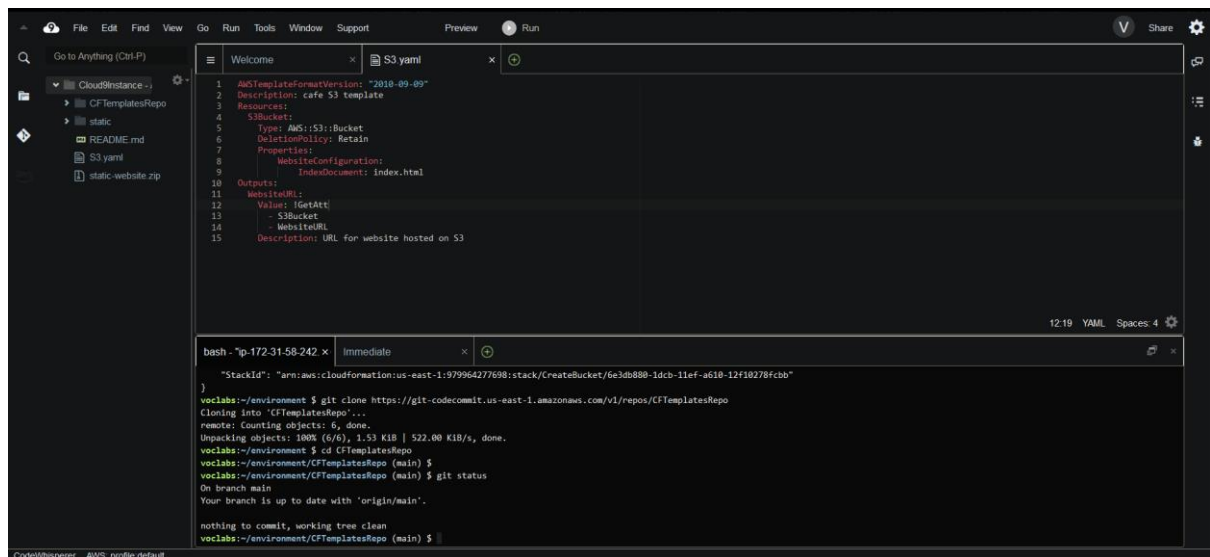


Task 3: Cloning a CodeCommit repository that contains AWS CloudFormation templates

In this task, I clone a CodeCommit repository to manage AWS CloudFormation templates for the café team. I start by navigating to the CodeCommit service and locating the CFTemplatesRepo repository. Inside, I explore the templates folder and examine the start-lab.yaml file, which defines resources like an AWS Cloud9 instance and the CodeCommit repository.



I copy the repository's HTTPS clone URL and return to the AWS Cloud9 IDE. In the Bash terminal, I run `git clone <url>` to clone the repository, creating a `CFTemplatesRepo` directory. I verify the clone by navigating to the directory and running `git status` to ensure everything is up to date.



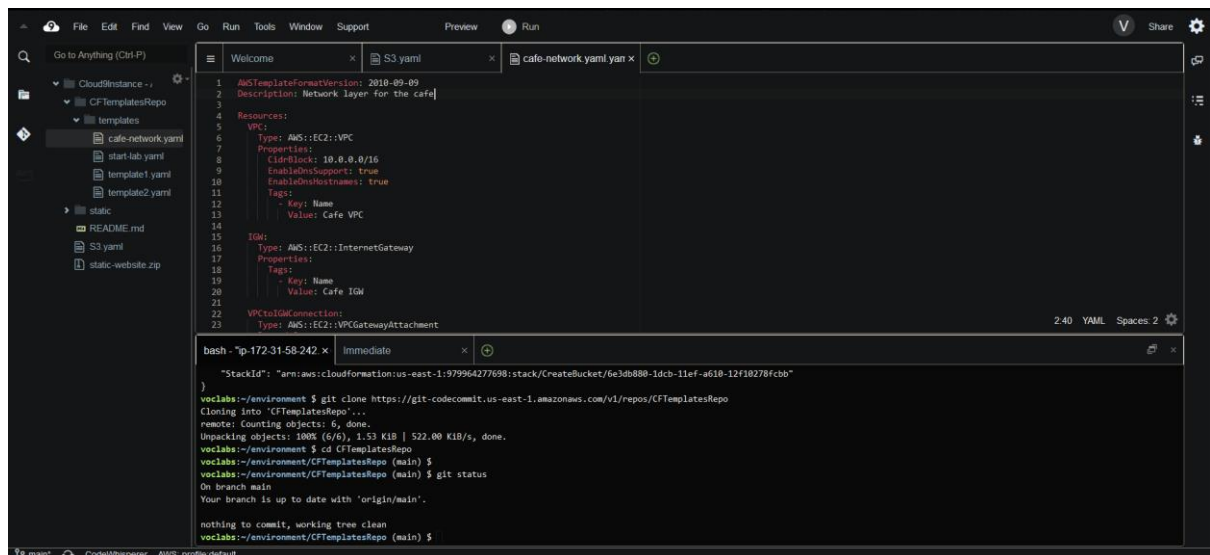
```
1 AWSTemplateFormatVersion: "2010-09-09"
2 Description: cafe S3 template
3 Resources:
4   S3Bucket:
5     Type: AWS::S3::Bucket
6     DeletionPolicy: Retain
7     Properties:
8       WebsiteConfiguration:
9         IndexDocument: index.html
10  Outputs:
11    WebsiteURL:
12      Value: !GetAtt
13        - S3Bucket
14        - WebsiteURL
15    Description: URL for website hosted on S3
```

```
bash - "ip-172-31-58-242" x Immediate x
"StackId": "arn:aws:cloudformation:us-east-1:979964277698:stack/CreateBucket/6e3db880-1dcb-11ef-a610-12f18278fcb"
}
voclabs:~/environment $ git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/CFTemplatesRepo
Cloning into 'CFTemplatesRepo'...
remote: Counting objects: 6, done.
Unpacking objects: 100% (6/6), 1.53 KiB | 522.00 KiB/s, done.
voclabs:~/environment $ cd CFTemplatesRepo
voclabs:~/environment/CFTemplatesRepo (main) $
voclabs:~/environment/CFTemplatesRepo (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
voclabs:~/environment/CFTemplatesRepo (main) $
```

Task 4: Creating a new network layer with AWS CloudFormation, CodeCommit, and CodePipeline

In this task, I create a VPC with a public subnet using an AWS CloudFormation template and deploy it via a CI/CD pipeline. I start by duplicating an existing template file in AWS Cloud9, renaming it cafe-network.yaml, and updating its description.

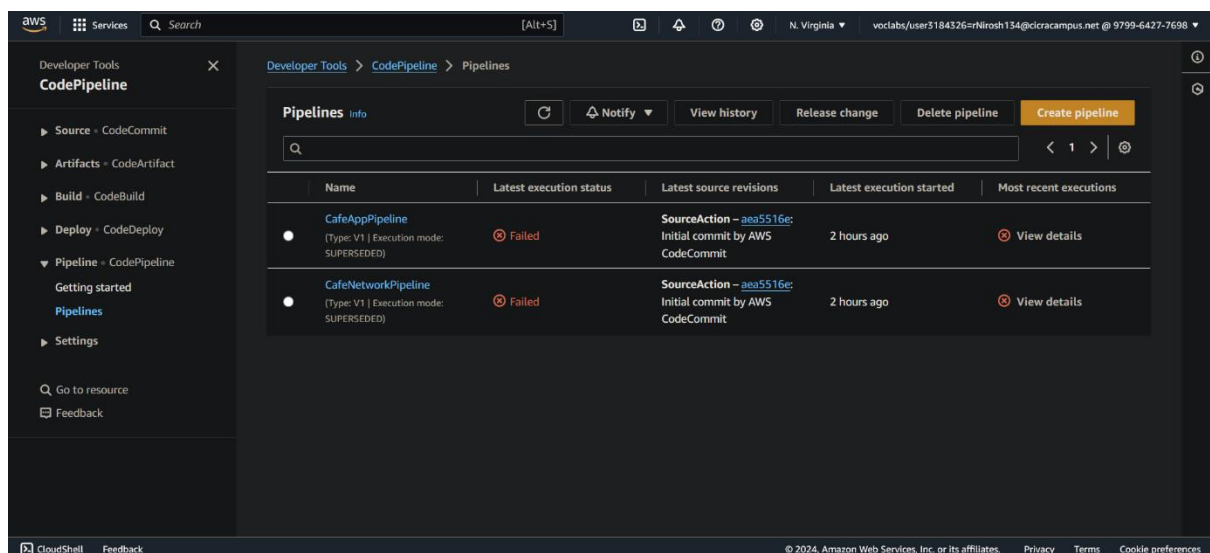


The screenshot shows the AWS Cloud9 IDE interface. The left sidebar displays the file explorer with a project structure including 'CloudInstance', 'CFTemplatesRepo', and 'static'. The main editor window shows the 'cafe-network.yaml' file with the following content:

```
1 AWSTemplateFormatVersion: 2010-09-09
2 Description: Network layer for the cafe
3
4 Resources:
5   VPC:
6     Type: AWS::EC2::VPC
7     Properties:
8       CidrBlock: 10.0.0.0/16
9       EnableDnsSupport: true
10      EnableDnsHostnames: true
11      Tags:
12        - Key: Name
13          Value: Cafe VPC
14
15   IGW:
16     Type: AWS::EC2::InternetGateway
17     Properties:
18       Tags:
19        - Key: Name
20          Value: Cafe IGW
21
22   VPCtoIGWConnection:
23     Type: AWS::EC2::VPCGatewayAttachment
```

The terminal window at the bottom shows the following commands and output:

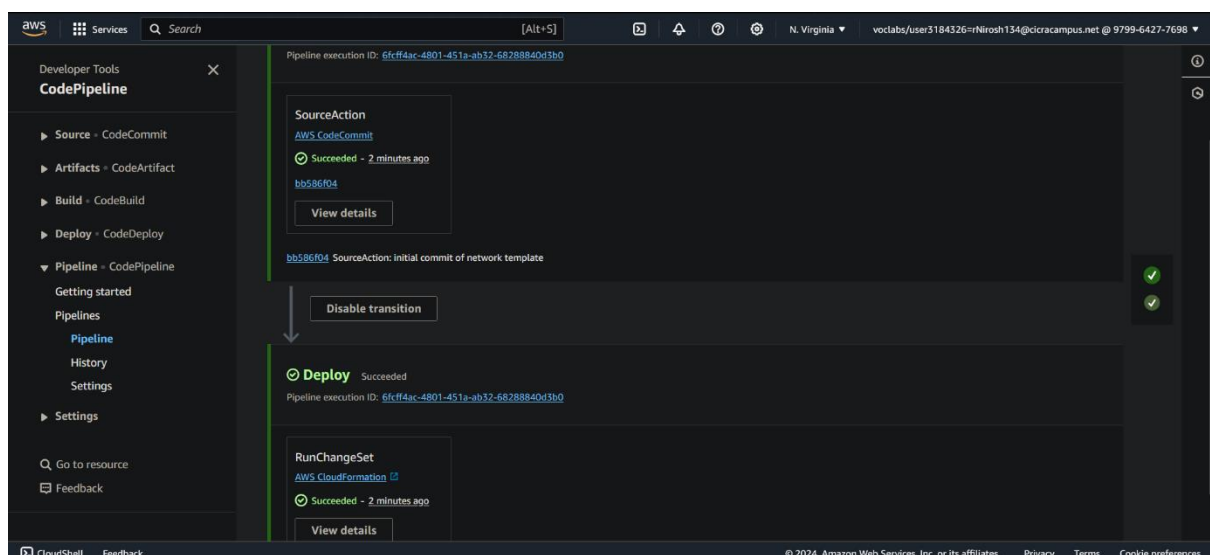
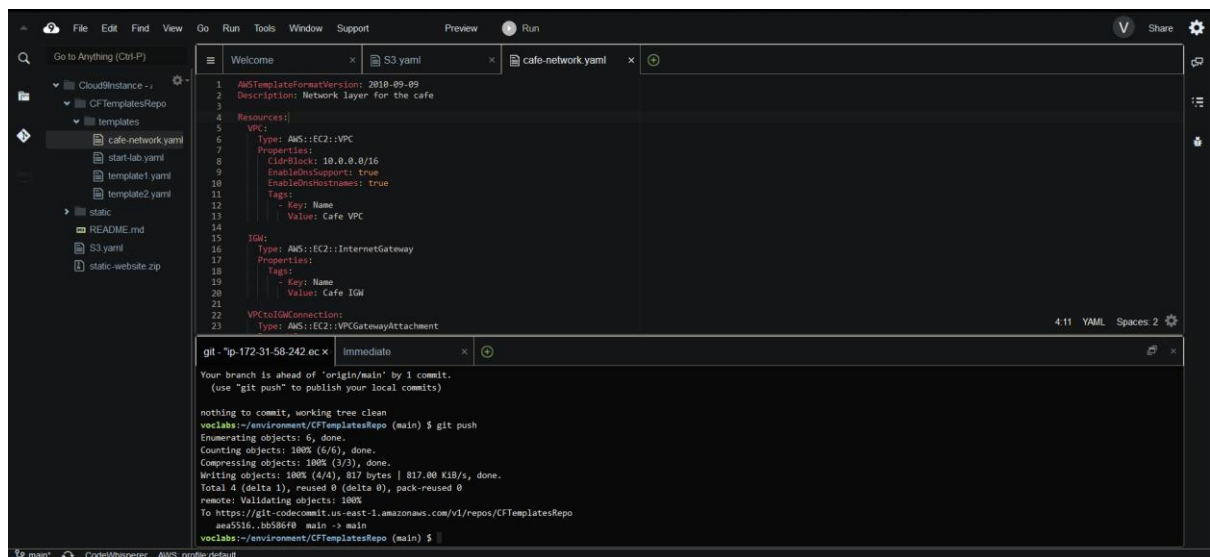
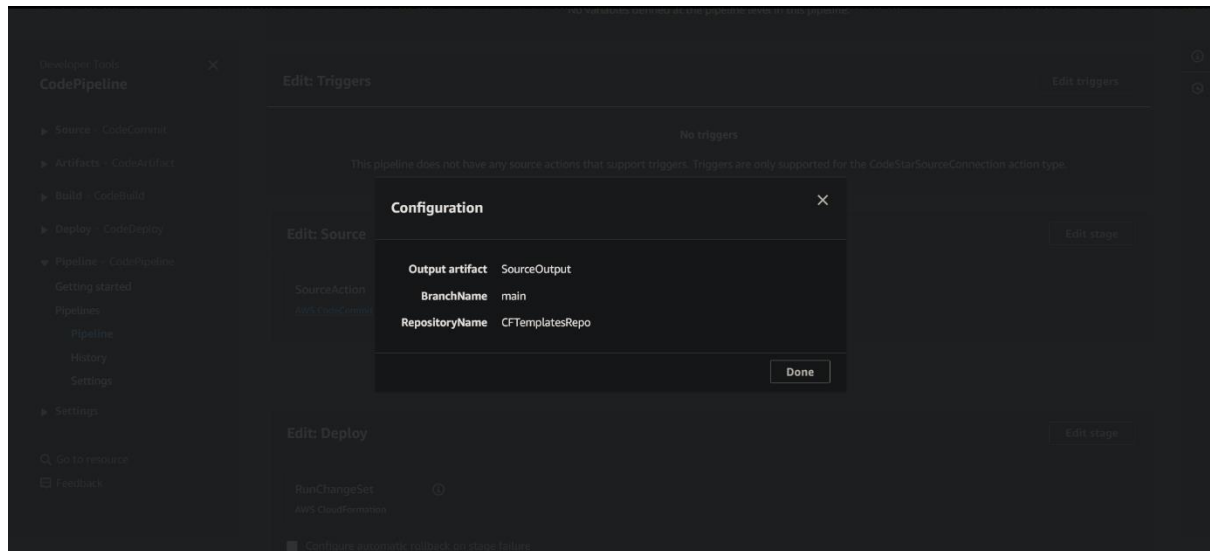
```
bash - "ip-172-31-58-242" x Immediate x
StackId: "arn:aws:cloudformation:us-east-1:979964277698:stack/CreateBucket/6e34b880-1dcb-11ef-a610-12f18278fcb"
}
voclabs:~/environment $ git clone https://git-codecommit.us-east-1.amazonaws.com/v1/repos/CFTemplatesRepo
Cloning into 'CFTemplatesRepo'...
remote: Counting objects: 6, done.
Unpacking objects: 100% (6/6), 1.53 KiB | 522.00 KiB/s, done.
voclabs:~/environment $ cd CFTemplatesRepo
voclabs:~/environment/CFTemplatesRepo (main) $
voclabs:~/environment/CFTemplatesRepo (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.
nothing to commit, working tree clean
voclabs:~/environment/CFTemplatesRepo (main) $
```



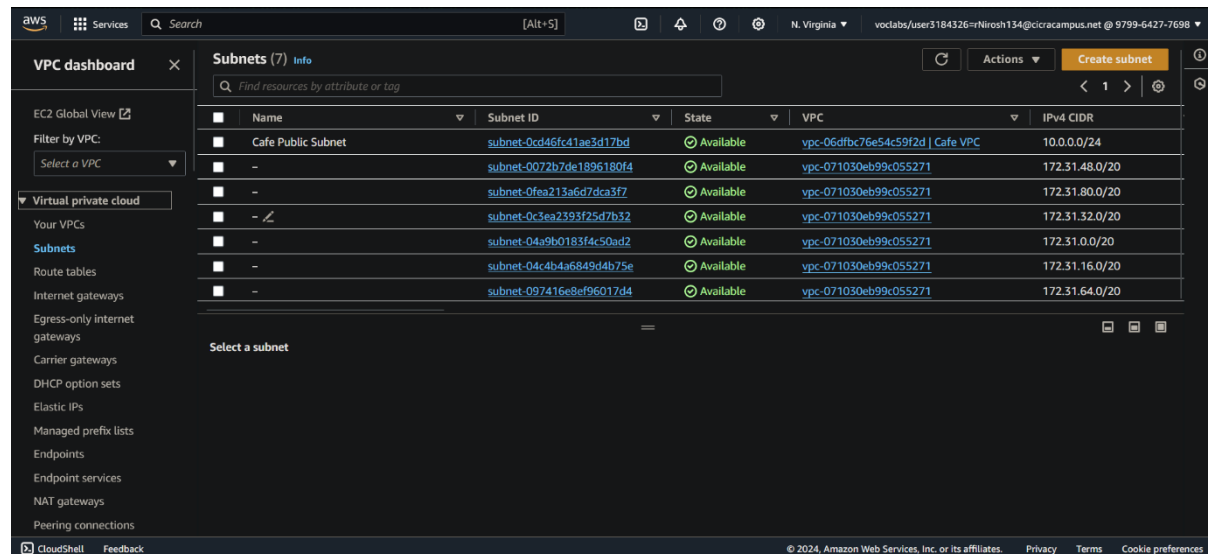
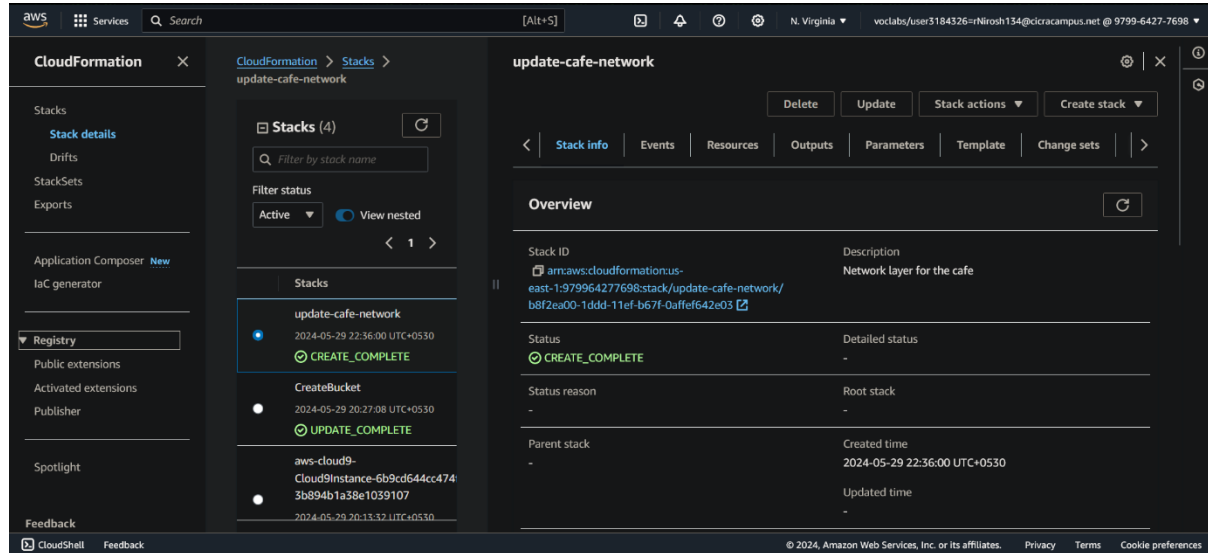
The screenshot shows the AWS CodePipeline console. The left sidebar contains the navigation menu with options like 'Source - CodeCommit', 'Artifacts - CodeArtifact', 'Build - CodeBuild', 'Deploy - CodeDeploy', and 'Pipeline - CodePipeline'. The main area displays the 'Pipelines' page with a table of existing pipelines.

Name	Latest execution status	Latest source revisions	Latest execution started	Most recent executions
CafeAppPipeline (Type: V1 Execution mode: SUPERSEDED)	Failed	SourceAction - aea5516e Initial commit by AWS CodeCommit	2 hours ago	View details
CafeNetworkPipeline (Type: V1 Execution mode: SUPERSEDED)	Failed	SourceAction - aea5516e Initial commit by AWS CodeCommit	2 hours ago	View details

I then check the preconfigured CafeNetworkPipeline in the AWS Management Console, noting that it references the CodeCommit repository. In Cloud9, I add, commit, and push the new template to CodeCommit, which triggers the pipeline. The pipeline creates or updates the update-cafe-network stack.

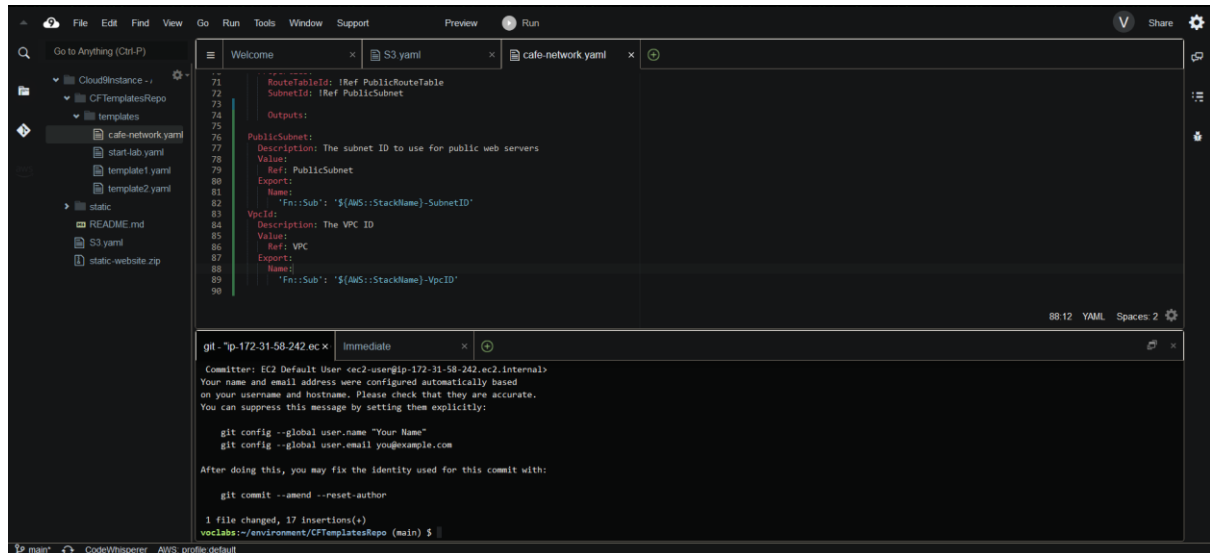


I verify the successful creation of the stack and network resources, such as the VPC and public subnet, in the AWS CloudFormation and VPC consoles.



Task 5: Updating the network stack

In Task 5, I updated the network stack by adding details about the resources it creates, like the public subnet and VPC ID, and ensured they could be referenced by other stacks.



```
71 RouteTableId: !Ref PublicRouteTable
72 SubnetId: !Ref PublicSubnet
73
74
75
76 Outputs:
77
78   PublicSubnet:
79     Description: The subnet ID to use for public web servers
80     Value:
81       Ref: PublicSubnet
82     Export:
83       Name:
84         Fn::Sub: '${AWS::StackName}-SubnetID'
85   VpcId:
86     Description: The VPC ID
87     Value:
88       Ref: VPC
89     Export:
90       Name:
91         Fn::Sub: '${AWS::StackName}-VpcID'
```

```
git - "ip-172-31-58-242.ec2" Immediate
Committer: EC2 Default User <ec2-user@ip-172-31-58-242.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

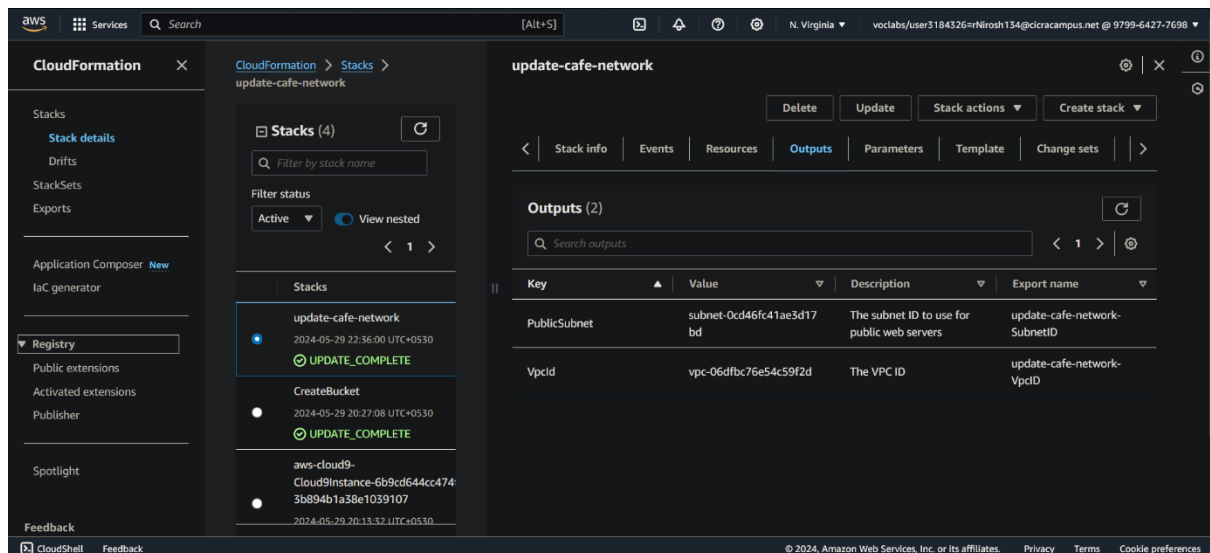
git config --global user.name "Your Name"
git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 17 insertions(+)
voclabs:~/environment/CFTemplatesRepo (main) $
```

After making these changes in the CloudFormation template, I committed them to CodeCommit using Git and verified that the stack update occurred successfully.



CloudFormation > **Stacks** > **update-cafe-network**

Stacks (4)

Filter by stack name

Filter status: Active View nested

Stacks

- update-cafe-network
2024-05-29 22:36:00 UTC+0530
UPDATE_COMPLETE
- CreateBucket
2024-05-29 20:27:08 UTC+0530
UPDATE_COMPLETE
- aws-cloud9-
Cloud9Instance-6b9cd644cc474:
3b894b1a38e1039107
2024-05-29 20:13:32 UTC+0530

update-cafe-network

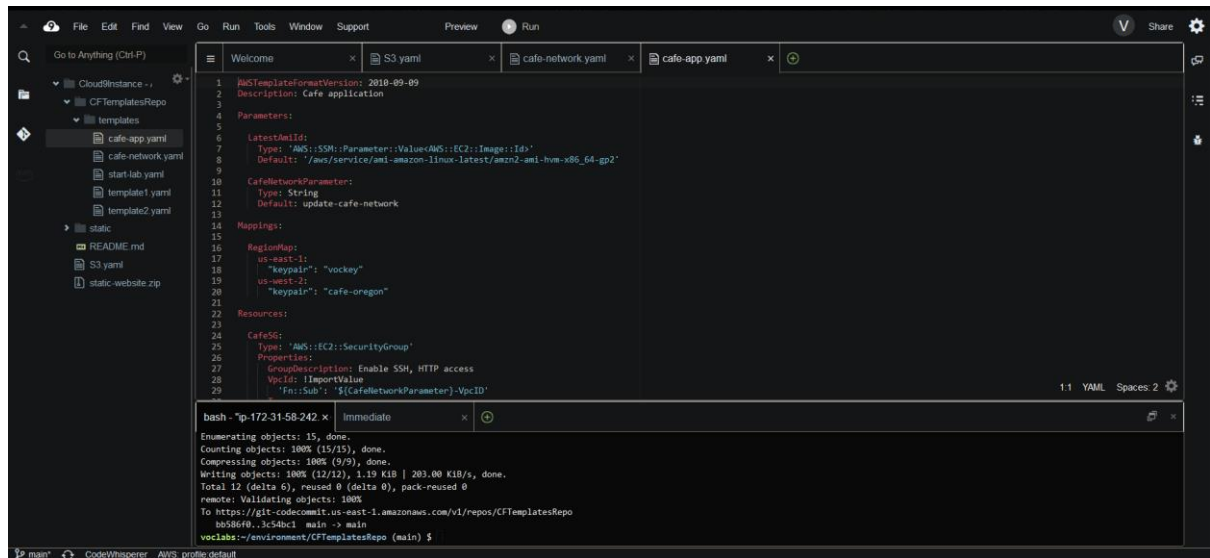
Stack info | Events | Resources | **Outputs** | Parameters | Template | Change sets

Outputs (2)

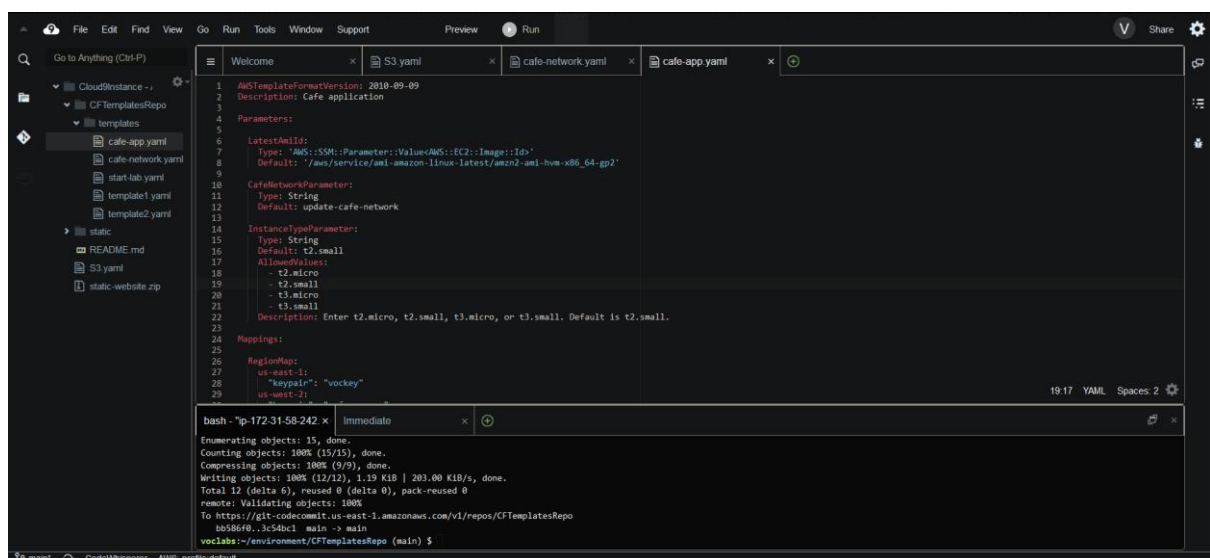
Key	Value	Description	Export name
PublicSubnet	subnet-0cd46fc41ae3d17bd	The subnet ID to use for public web servers	update-cafe-network-SubnetID
VpcId	vpc-06dfbc76e54c59f2d	The VPC ID	update-cafe-network-VpcID

Task 6: Defining an EC2 instance resource and creating the application stack

Moving to Task 6, I created a new CloudFormation template to deploy an EC2 instance for the café website. After duplicating an existing template and analyzing its contents, I defined a new parameter to allow users to choose different instance types for the EC2 instance.

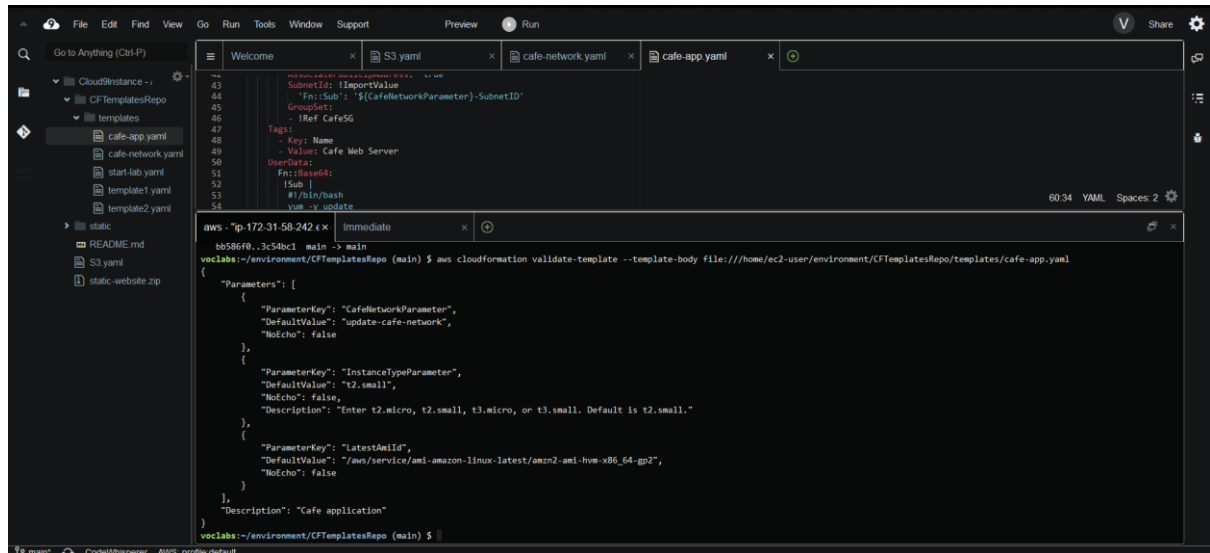


```
1 AWSTemplateFormatVersion: 2010-09-09
2 Description: Cafe application
3
4 Parameters:
5
6 LatestAmiId:
7   Type: 'AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>'
8   Default: '/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2'
9
10 CafeNetworkParameter:
11   Type: String
12   Default: update-cafe-network
13
14 Mappings:
15
16 RegionMap:
17   us-east-1:
18     "keypair": "vockey"
19   us-west-2:
20     "keypair": "cafe-oregon"
21
22 Resources:
23
24 CafeSG:
25   Type: 'AWS::EC2::SecurityGroup'
26   Properties:
27     GroupDescription: Enable SSH, HTTP access
28     VpcId: !ImportValue
29     'Fn::Sub': '${CafeNetworkParameter}-VpcID'
```



```
1 AWSTemplateFormatVersion: 2010-09-09
2 Description: Cafe application
3
4 Parameters:
5
6 LatestAmiId:
7   Type: 'AWS::SSM::Parameter::Value<AWS::EC2::Image::Id>'
8   Default: '/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2'
9
10 CafeNetworkParameter:
11   Type: String
12   Default: update-cafe-network
13
14 InstanceTypeParameter:
15   Type: String
16   Default: t2.small
17   AllowedValues:
18     - t2.micro
19     - t2.small
20     - t3.micro
21     - t3.small
22   Description: Enter t2.micro, t2.small, t3.micro, or t3.small. Default is t2.small.
23
24 Mappings:
25
26 RegionMap:
27   us-east-1:
28     "keypair": "vockey"
29   us-west-2:
30     "keypair": "cafe-oregon"
```

Following the AWS Documentation as a guide, I then created the EC2 instance resource with specific characteristics, such as its logical ID, image ID, and instance type.

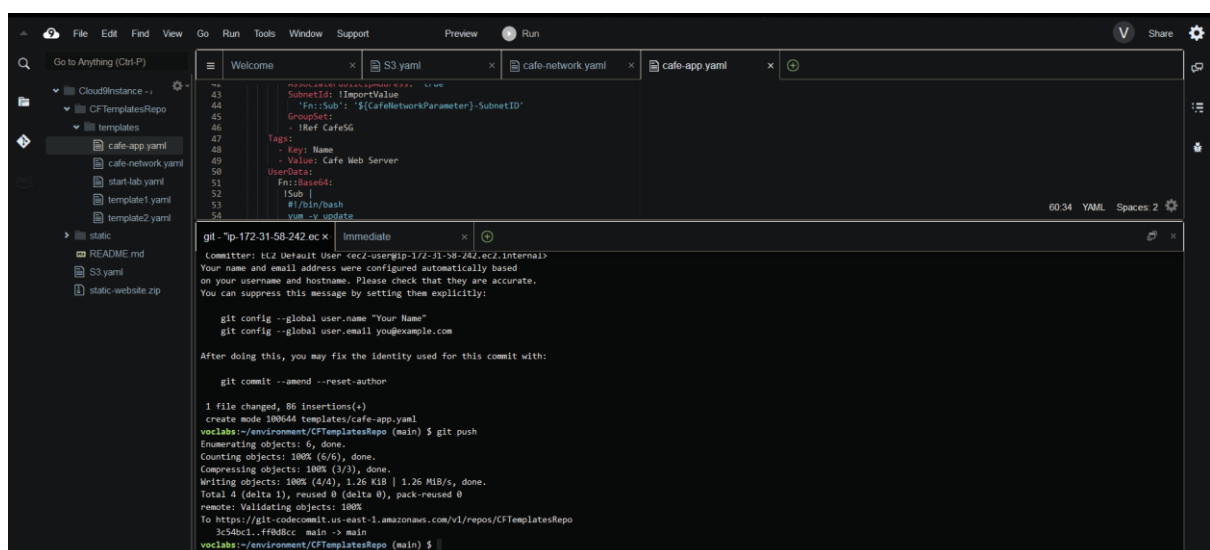


The screenshot shows a code editor with a file explorer on the left and a terminal at the bottom. The file explorer shows a project structure with folders like 'CloudInstance', 'CF-TemplatesRepo', and 'templates'. The 'templates' folder contains files like 'cafe-app.yaml', 'cafe-network.yaml', 'start-lab.yaml', 'template1.yaml', and 'template2.yaml'. The 'cafe-app.yaml' file is open in the editor, showing a CloudFormation template snippet. The terminal shows the command 'aws cloudformation validate-template' being executed, and the output indicates that the template is valid.

```
43 SubnetId: !ImportValue
44   Fn::Sub: ${CafeNetworkParameter}-SubnetID
45 GroupSet:
46   - !Ref CafeSG
47 Tags:
48   - Key: Name
49     Value: Cafe Web Server
50 UserData:
51   Fn::Base64:
52     |Sub|
53     #!/bin/bash
54     yum -y update
```

```
aws - "p-172-31-58-242" x Immediate
bb586f0..3c54bc1 main -> main
voelabs:~/environment/CF-TemplatesRepo (main) $ aws cloudformation validate-template --template-body file:///home/ec2-user/environment/CF-TemplatesRepo/templates/cafe-app.yaml
{
  "Parameters": [
    {
      "ParameterKey": "CafeNetworkParameter",
      "DefaultValue": "update-cafe-network",
      "NoEcho": false
    },
    {
      "ParameterKey": "InstanceTypeParameter",
      "DefaultValue": "t2.small",
      "NoEcho": false,
      "Description": "Enter t2.micro, t2.small, t3.micro, or t3.small. Default is t2.small."
    },
    {
      "ParameterKey": "LatestAmiId",
      "DefaultValue": "/aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2",
      "NoEcho": false
    }
  ],
  "Description": "Cafe application"
}
voelabs:~/environment/CF-TemplatesRepo (main) $
```

After validating the template's format, I committed and pushed the changes to CodeCommit and monitored the progress in the CodePipeline console.



The screenshot shows the same code editor as before, but the terminal now displays the output of a git commit and push command. The output shows that the commit was successful and the code was pushed to the repository.

```
git - "p-172-31-58-242" oc x Immediate
Commit: tLZ Default User <ec2-user@ip-172-31-58-242.ec2.internal>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

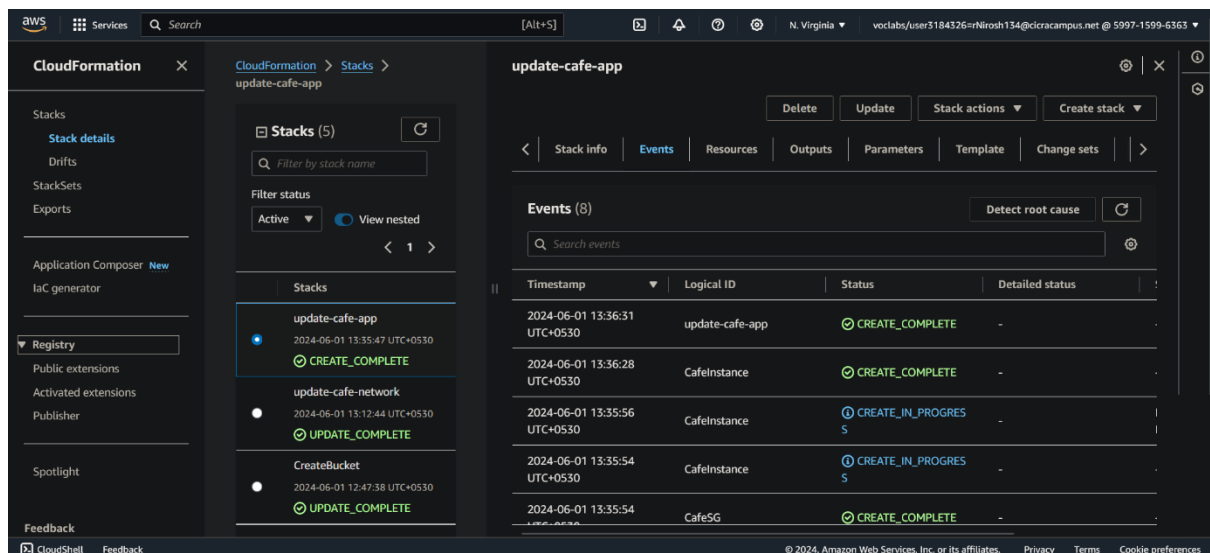
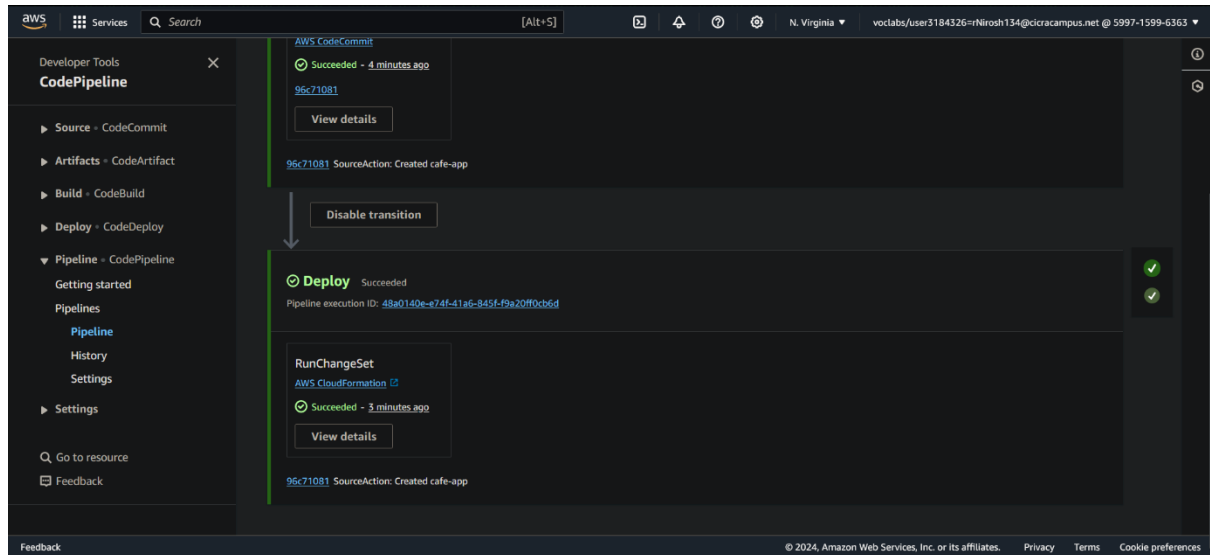
git config --global user.name "Your Name"
git config --global user.email you@example.com

After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

1 file changed, 86 insertions(+)
create mode 100644 templates/cafe-app.yaml
voelabs:~/environment/CF-TemplatesRepo (main) $ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.25 KiB | 1.25 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/CF-TemplatesRepo
3c54bc1..ff0d8cc main -> main
voelabs:~/environment/CF-TemplatesRepo (main) $
```

Once the deployment was successful, I verified the creation of the EC2 instance in the Amazon EC2 console and tested the café website using its public IPv4 address.

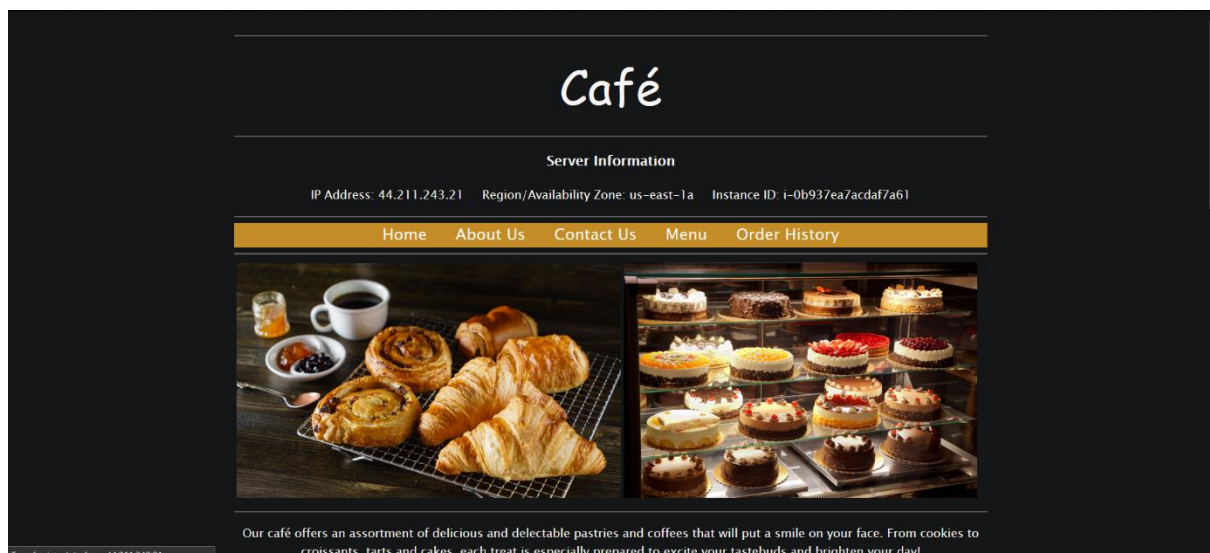
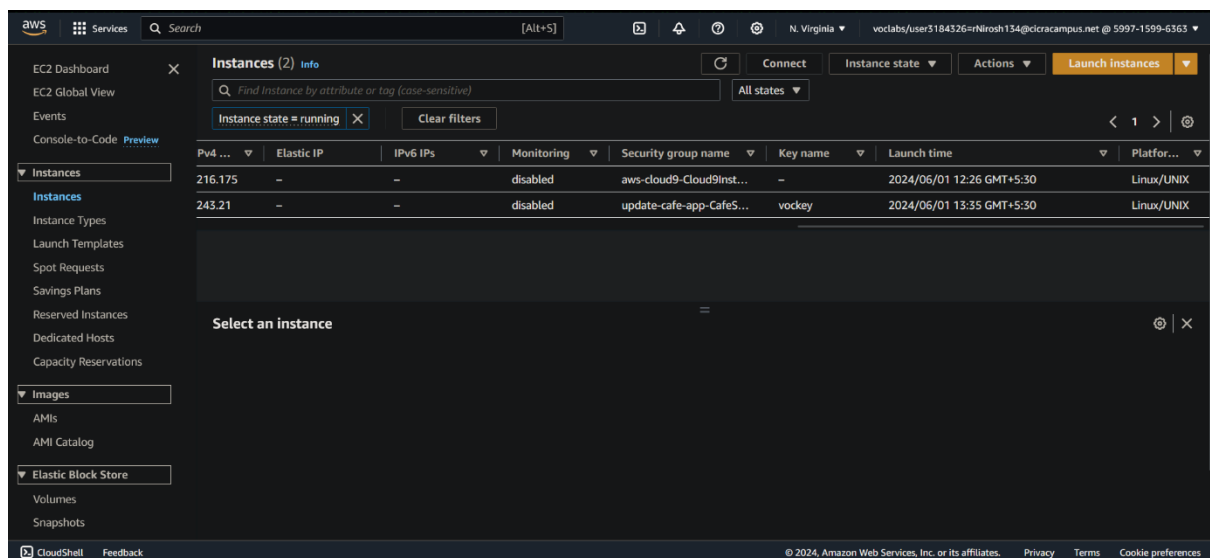
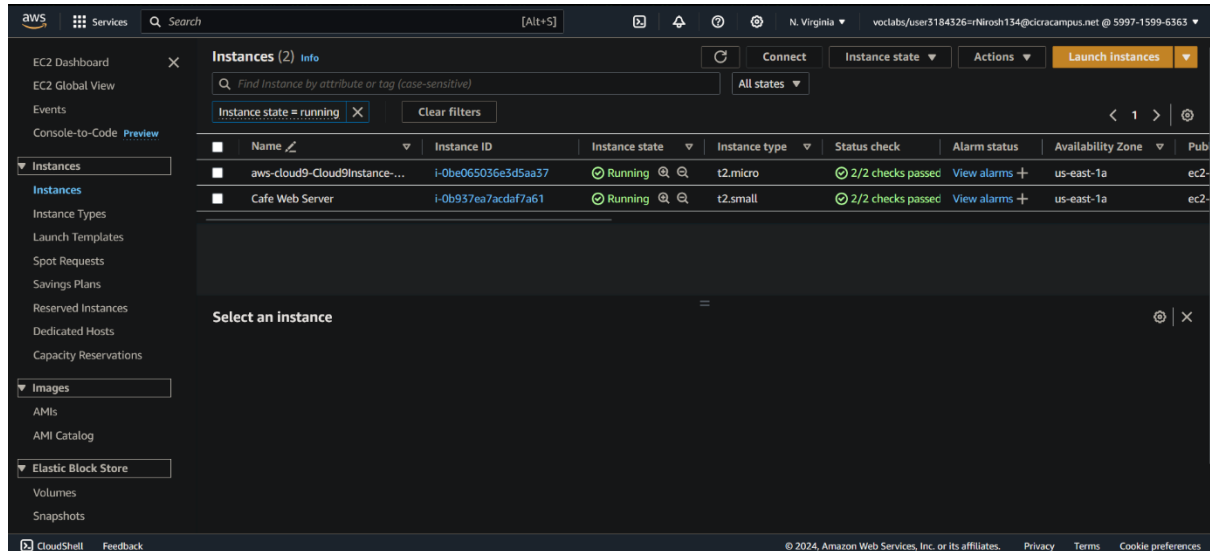


SIT233 – Cloud Computing

Module 10 Challenge Lab - Automating Infrastructure Deployment

Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Through these tasks, I effectively updated the network stack and deployed the application stack using a CI/CD pipeline and Infrastructure as Code approach.



Task 7: Duplicating the café network and website to another AWS Region

Question 4: Go to the Parameters tab of the update-cafe-app stack. What value do you see for the LatestAmiId?

- ☐ /aws/service/ami-windows-latest
- ☐ /aws/service/ami-amazon-linux-latest/amzn-ami-minimal-hvm-x86_64-ebs
- ☐ The value is blank
- ☒ /aws/service/ami-amazon-linux-latest/amzn2-ami-hvm-x86_64-gp2

Submit

Question 5: Go to the Stack info tab of the update-cafe-app stack. What is the Amazon Resource Name (ARN) of the IAM role that grants the permissions to create and update the update-cafe-app stack?

- ☒ CustomCFnRole
- ☐ MyCloudFormationRole
- ☐ AWSCloudFormationReadOnlyAccess
- ☐ AWSCloudFormationFullAccess

Submit

Question 6: In the AWS Management Console, navigate to the CodeCommit repository where your AWS CloudFormation templates are stored. Choose Commits and in the Commits list, open one of the commits by choosing its commit ID. What do you observe?

- ☐ A listing of the files affected by the commit.
- ☐ The timestamp of when the commit was made and by whom.
- ☒ Details including what lines of which files were changed as part of the commit and what the lines now contain.
- ☐ I did not have permission to view the commit details.

Submit

In Task 7, I duplicated the café network resources in another AWS Region using the AWS CLI and AWS CloudFormation console. First, I used the AWS CLI to create a stack in the specified Region, which returned a StackId.

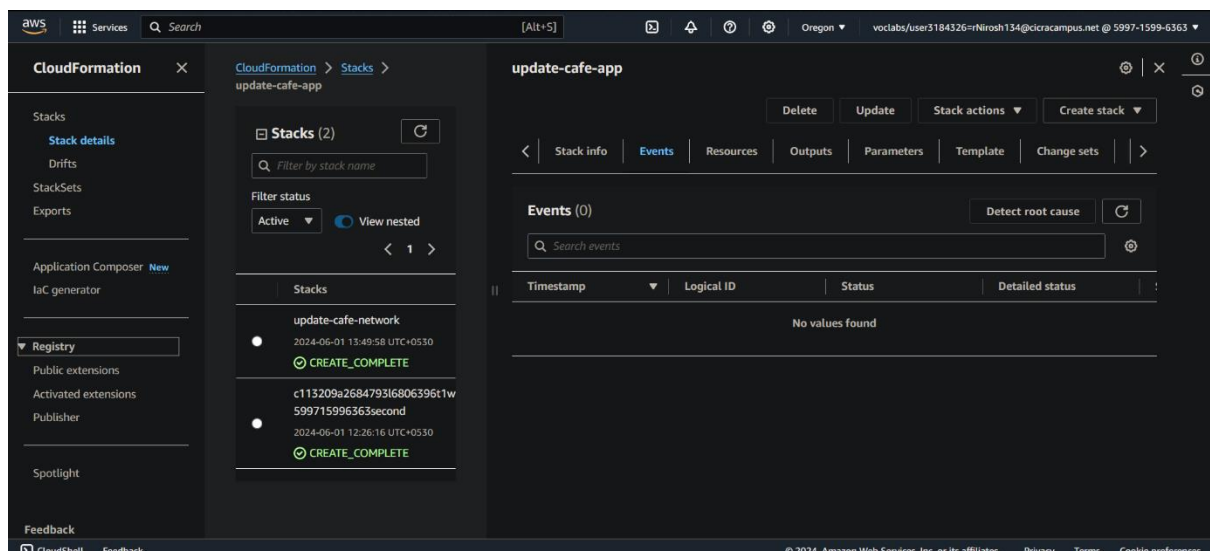
```
aws - "p-172-31-12-118 ex" Immediate x
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly:

git config --global user.name "Your Name"
git config --global user.email you@example.com

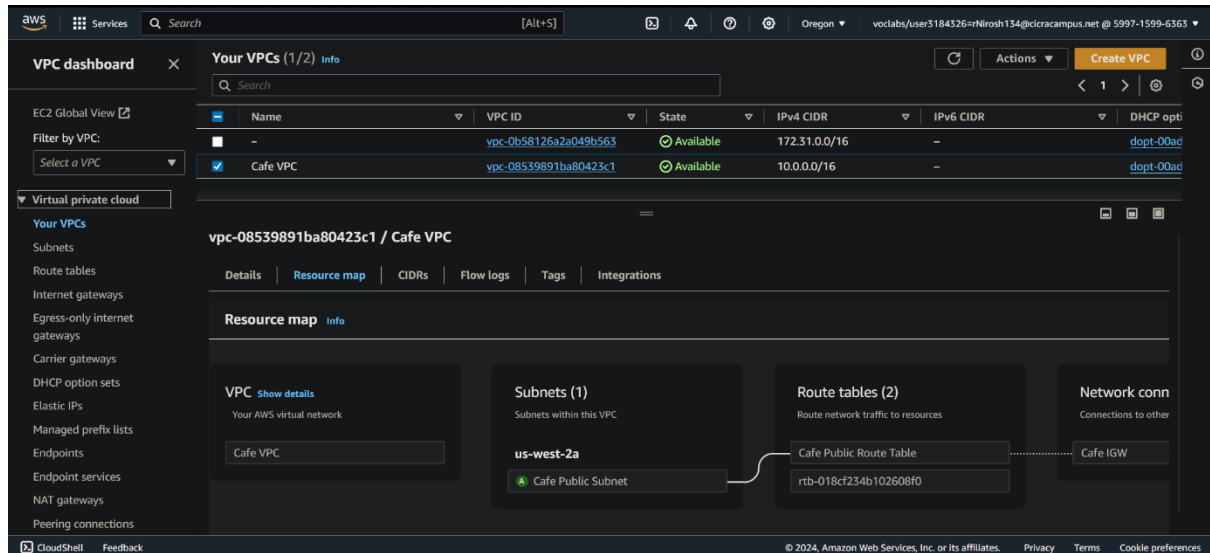
After doing this, you may fix the identity used for this commit with:

git commit --amend --reset-author

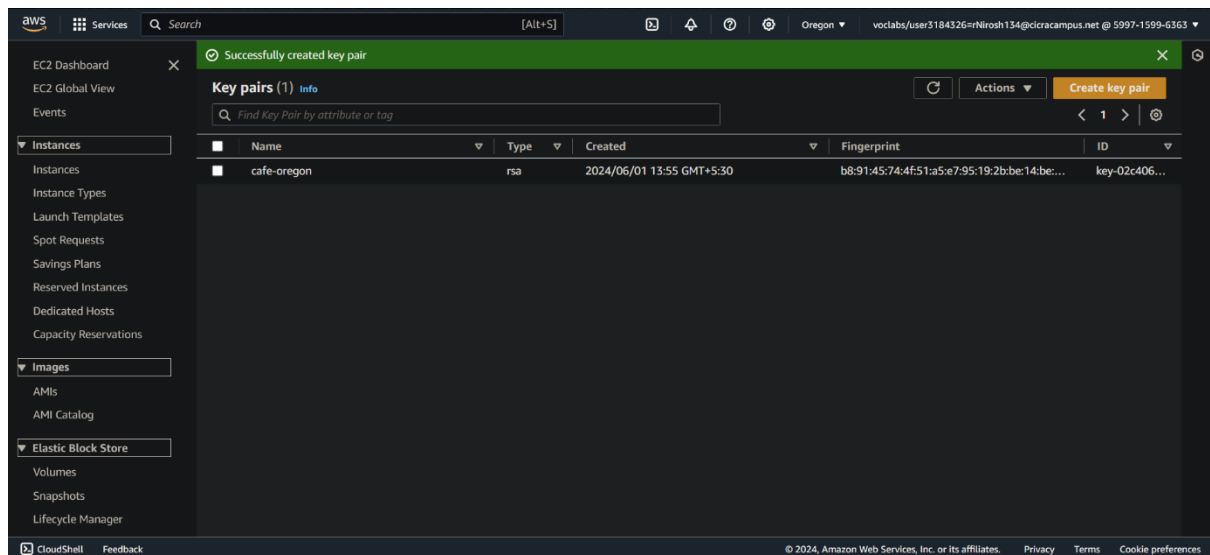
1 file changed, 88 insertions(+)
create mode 100644 templates/cafe-app.yaml
voclabs:~/environment/CFTemplatesRepo (main) $ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.25 KiB | 1.25 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/CFTemplatesRepo
3b6c7108 main -> main
voclabs:~/environment/CFTemplatesRepo (main) $ aws cloudformation create-stack --stack-name update-cafe-network --template-body file:///home/ec2-user/environment/CFTemplatesRepo/templates/cafe-network.yaml --region us-west-2
{
  "StackId": "arn:aws:cloudformation:us-west-2:599715996363:stack/update-cafe-network/bbc1da0-1fef-11ef-ale4-8a7787995e7"
}
voclabs:~/environment/CFTemplatesRepo (main) $
```



Then, I switched to the AWS CloudFormation console in the new Region and verified that the stack's status changed to CREATE_COMPLETE.



After confirming the creation of network resources in the new Region, I created an EC2 key pair named cafe-oregon and revisited the application template details in the Cloud9 IDE.



The screenshot shows the AWS CloudShell terminal with the following content:

```

aws -p-172-31-12-118 ex Immediate x
git config --global user.name "Your Name"
git config --global user.email you@example.com

After doing this, you may fix the Identity used for this commit with:

git commit --amend --reset-author

1 file changed, 88 insertions(+)
create mode 100644 templates/cafe-app.yaml
voclabs:~/environment/CFTemplatesRepo (main) $ git push
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 1.25 KiB | 1.25 MiB/s, done.
Total 4 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Validating objects: 100%
To https://git-codecommit.us-east-1.amazonaws.com/v1/repos/CFTemplatesRepo
   96c7108 main -> main
voclabs:~/environment/CFTemplatesRepo (main) $ aws cloudformation create-stack --stack-name update-cafe-network --template-body file:///home/ec2-user/environment/CFTemplatesRepo/templates/cafe-network.yaml --region us-west-2
{
  "StackId": "arn:aws:cloudformation:us-west-2:599715996363:stack/update-cafe-network/bc1cda0-1fef-11ef-a1e4-8a78799e5e7"
}
voclabs:~/environment/CFTemplatesRepo (main) $ aws s3 cp templates/cafe-app.yaml s3://c113209a2684793168863961w599715996363-repobucket-wdxtcarg2cu/
upload: templates/cafe-app.yaml to s3://c113209a2684793168863961w599715996363-repobucket-wdxtcarg2cu/cafe-app.yaml
voclabs:~/environment/CFTemplatesRepo (main) $
  
```

The left sidebar shows the file explorer with the following structure:

- CloudInstance -> CFTemplatesRepo -> templates -> cafe-app.yaml
- cafe-network.yaml
- start-lab.yaml
- template1.yaml
- template2.yaml
- static -> README.md
- S3.yaml
- static-website.zip

aws

Services

Search

[Alt+S]

📧

🔔

🔄

🌐

Oregon

voclabs/user3184326@nrosh134@cdcracampus.net @ 5997-1599-6363

☰

CloudFormation > Stacks > cafe-app

Stacks (3)

Filter status

Filter by stack name

Active

View nested

Stacks

cafe-app

2024-06-01 14:03:55 UTC+0530

CREATE_COMPLETE

update-cafe-network

2024-06-01 13:49:58 UTC+0530

CREATE_COMPLETE

c113209a268479316806396t1w599715996363second

2024-06-01 12:26:16 UTC+0530

CREATE_COMPLETE

cafe-app

Stack info | Events | Resources | Outputs | Parameters | Template | Change sets | Git

Overview

Stack ID

arn:aws:cloudformation:us-west-2:599715996363:stack/cafe-app/aee69a0-1ff1-11ef-b14f-02c23e8ee57b

Description

Cafe application

Status

CREATE_COMPLETE

Detailed status

-

Status reason

-

Root stack

-

Parent stack

-

Created time

2024-06-01 14:03:55 UTC+0530

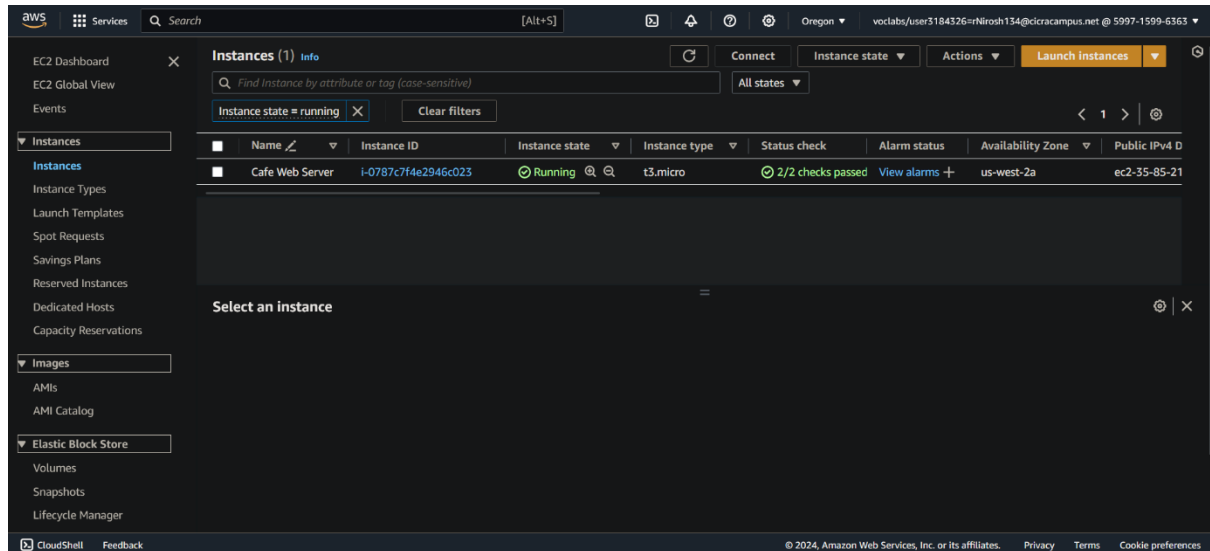
Updated time

-

Deleted time

Drift status

Finally, I verified the creation of resources in the Amazon EC2 console and accessed the café website using the public IPv4 IP address of the EC2 instance, noting the server information indicating the new Region.



Through this process, I efficiently duplicated the café network and website to another AWS Region.



SIT233 – Cloud Computing
Module 10 Challenge Lab - Automating Infrastructure
Deployment

Name: Nirosh Ravindran
CICRA Student ID: BSCP|CS|63|134

Grade:

My grade is 30/30

The screenshot shows the AWS Academy interface for a challenge lab. The left sidebar contains navigation links: Account, Dashboard, Courses, Calendar, Inbox, History, and Help. The main content area is titled 'Module 10 Challenge Lab - Automating Infrastructure Deployment'. Below the title, it shows 'Due No Due Date', 'Points 100', and 'Submitting an external tool'. A toolbar includes buttons for 'Submit', 'Details', 'AWS', 'Start Lab', 'End Lab', '1:16', 'Instructions', 'Grades', and 'Actions'. The main workspace is divided into three sections: a left pane with a dropdown menu showing 'EN_US' and the text 'the us-east-1 Region.', a central pane with the heading 'Update from the café' and a photo of two people, and a right pane with a 'Launch Terminal' button. On the far right, a 'Grades' panel shows a 'Total score' of 30/30 and a list of five answers, each with a score of 1/1.

Files	README	Terminal	Source
	☒	☒	☐

Answer	Score
[Answer 01]	1/1
[Answer 02]	1/1
[Answer 03]	1/1
[Answer 04]	1/1
[Answer 05]	1/1